

Jose Manuel Hernandez de Labra

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EDUCATION

B.Sc. in Biomedical Engineering with a Minor in Neuroscience <i>Tecnologico de Monterrey (ITESM)</i>	Monterrey, NL, Mexico Aug. 2018 – Dec 2023
M.Sc. in Bioinformatics for Health Sciences <i>Jointly organized by <u>Universitat Pompeu Fabra (UPF)</u> and the <u>University of Barcelona (UB)</u></i>	Barcelona, Spain Sep. 2025 – Present

EXPERIENCE

Neuroscience Department, Zambrano Hospital <i>Data Scientist Intern</i>	Aug. 2022 – Dec. 2022 Monterrey, NL
- Cleaned and processed research data collected by psychologists and psychiatrists. - Developed interactive data visualizations, enabling better comprehension of study findings. - Applied machine learning models for patient classification and anomaly detection, improving diagnostic precision.	
Core Lab Genomics <i>Machine Learning Researcher</i>	Jan. 2024 – May. 2025 Monterrey, NL
- Designed and optimized machine learning architectures for RNA and biomolecule analysis. - Predicted 3D RNA structures, contributing to advancements in molecular interaction studies. - Applied advanced docking algorithms to predict biomolecule-protein interactions, aiding in therapeutic target discovery.	

RESEARCH PROJECTS

rnAIR <i>Python, Tensorflow, Molecular Biology, Mathematical Modeling</i> <i>IGEM competition</i>	Jan. 2022 – Dec. 2022
- Developed neural network models to predict small RNA binding sites, enabling bacterial gene inhibition. - Collected and analyzed extensive datasets of small RNAs, addressing antibiotic resistance issues. - Contributed to mathematical modeling and programming efforts, ensuring model robustness and accuracy.	
ADA <i>Python, Docker, Tensorflow, Structural Bioinformatics, Molecular Dynamics</i> <i>IGEM Design League</i>	Jan. 2023 – Dec. 2023
- Led mathematical modeling and programming efforts to develop biosensors for Alzheimer's detection. - Designed aptamers using 3D RNA models to bind specific proteins. - Trained deep learning architectures, including mesh convolutions and GAN's, on Azure Virtual Machines.	
STRAND <i>Python, Azure, Pytorch, Molecular Dynamics</i> <i>Tecnologico de Monterrey in collaboration with Microsoft AI For Good Lab</i>	Jan. 2024 – Jan. 2025
- Conducted research to improve RNA 3D structure prediction, enhancing insights into RNA folding and interactions. - Generated labeled datasets with precise intramolecular bindings. - Developed graph neural networks for improved secondary structure contact maps. - Utilized contact maps as templates for accurate 3D RNA structure prediction.	

TECHNICAL SKILLS

Programming Languages: Python, Assembler, Bash, SQL, C, C++, CUDA LaTeX
Developer Tools: Git, Docker, Azure, Conda
ML Frameworks: Tensorflow, Pytorch, SkLearn, PyTorch Geometric
Languages: Spanish - (Native), **English** - (C1), **French** - (A1)

ADDITIONAL INFORMATION

Awards and Achievements
Two Gold Medals, iGEM Competition (2022, 2023) – Recognized for excellence in synthetic biology research and innovative solutions. Judge, iGEM Competition (2024, 2025) – Selected to evaluate and mentor competing teams, showcasing leadership and subject matter expertise.